

4

The z-Transform

Objectives

- Students understand the z-transform and its applications for the discrete-time signals and systems.

Why Do We Need The z-Transform?

- The Discrete-time Fourier Transform is useful for operations with steady-state, e.g., sinusoidal, signals in the frequency –domain.
- We need another tool for signals like $u(n)$ or $nu(n)$.
- The z-transform can do what DTFT does with extended capabilities in analysis of the discrete-time signals and systems.

The z-Transform of discrete-time signal $x(n)$

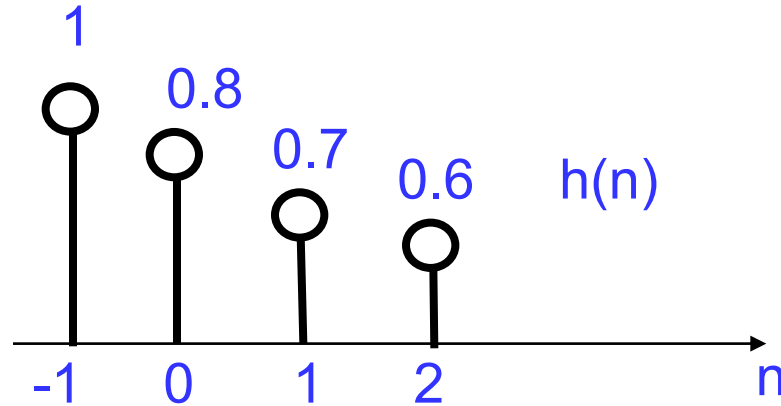
- The z-transform

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

- The z-transform pair

$$x(n) \xleftrightarrow{z} X(z)$$

Example 4.1



$$h(n) = \{1, 0.8, 0.7, 0.6\}$$

↑

$$H(z) = z + 0.8 + 0.7z^{-1} + 0.6z^{-2}$$

Fourier transform vs. z-transform

- From $z = re^{j\omega}$, we have

$$\begin{aligned} X(re^{j\omega}) &= \sum_{n=-\infty}^{\infty} x(n)(re^{j\omega})^{-n} \\ &= \sum_{n=-\infty}^{\infty} (x(n)r^{-n})e^{-j\omega n} \end{aligned}$$

- Which is the Fourier transform of $x(n)r^{-n}$

$$X(re^{j\omega}) = F\{x(n)r^{-n}\}$$

**For $r=1$,
the z-transform becomes
the Fourier transform**

$$X(z)\big|_{z=e^{j\omega}} = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} = X(e^{j\omega})$$

Example 4.2

- Consider

$$x(n) = a^n u(n)$$

- Determine the z-transform of $x(n)$.

The z-transform of $x(n) = a^n u(n)$

$$X(z) = \sum_{n=-\infty}^{\infty} (a^n u(n)) z^{-n}$$

Remember the infinite geometric formula

$$= \sum_{n=0}^{\infty} a^n z^{-n}$$

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \quad a < 1$$

$$= \sum_{n=0}^{\infty} (az^{-1})^n$$

$$= \frac{1}{1-az^{-1}}, \quad |az^{-1}| < 1 \quad \text{or} \quad |z| > |a|$$

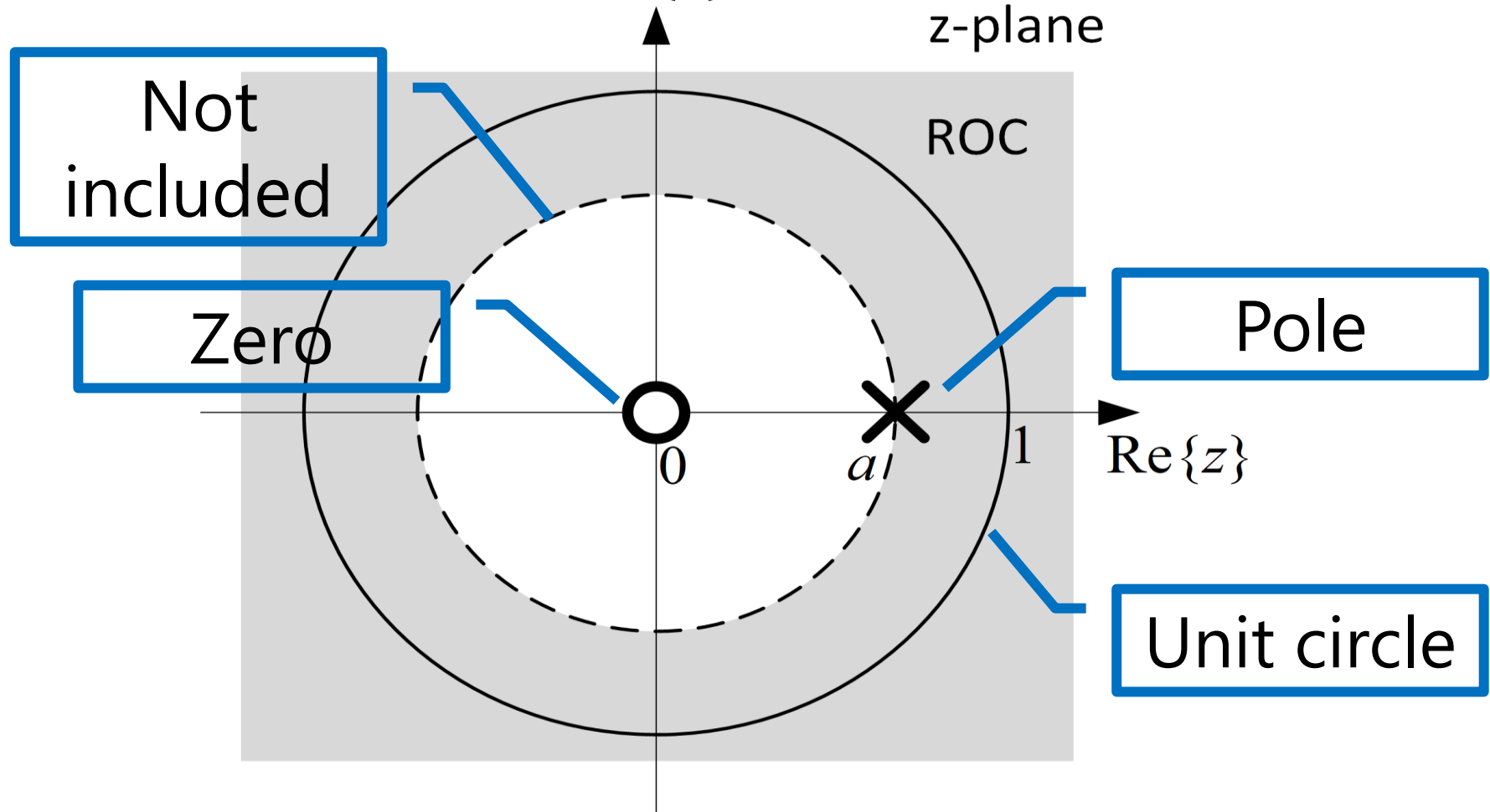
The Region of Convergence (ROC)

- For the series to converge, we need the condition

$$\left| az^{-1} \right| < 1 \quad \text{or} \quad |z| > |a|$$

The region of z that corresponds to the condition above is called **the Region of Convergence (ROC)** of the z -transform.

$$X(z) = \frac{1}{1 - az^{-1}}, \quad |z| > |a|$$



A z-Transform pair

- So the solution of the z-transform is then

$$X(z) = \frac{1}{1 - az^{-1}}, \quad |z| > |a|$$

- For this example, we express the z-transform pair as

$$a^n u(n) \xleftrightarrow{z} \frac{1}{1 - az^{-1}}, \quad |z| > |a|$$

Causal Systems

- A system is **causal, i.e.**, a system where the output depends on past and current inputs but not future inputs, iff

$$h(n) = 0, \quad n < 0$$

- Then $x(n) = a^n u(n)$ is a causal system.

Example 4.3

- Determine the z-transform of

$$x(n) = -a^n u(-n - 1)$$

- Note that if $x(n)$ is the impulse response of a system, this system is **non-causal** since $x(n)$ has non-zero components when $n < 0$.

$$\begin{aligned}
X(z) &= \sum_{n=-\infty}^{\infty} (-a^n u(-n-1)) z^{-n} = - \sum_{n=-\infty}^{-1} a^n z^{-n} \\
&= 1 - \sum_{n=-\infty}^0 (az^{-1})^n = 1 - \sum_{n=0}^{\infty} (a^{-1}z)^n \\
&= 1 - \frac{1}{1 - a^{-1}z}, \quad |a^{-1}z| < 1 \\
&= \frac{1}{1 - az^{-1}}, \quad |a^{-1}z| < 1
\end{aligned}$$

The Region of Convergence (ROC)

- For the series to converge $|a^{-1}z| < 1$ or

$$|z| < |a|$$

- Is the ROC of the z-transform of $x[n] = -a^n u(-n-1)$

The z-transform of $x(n) = -a^n u(-n - 1)$

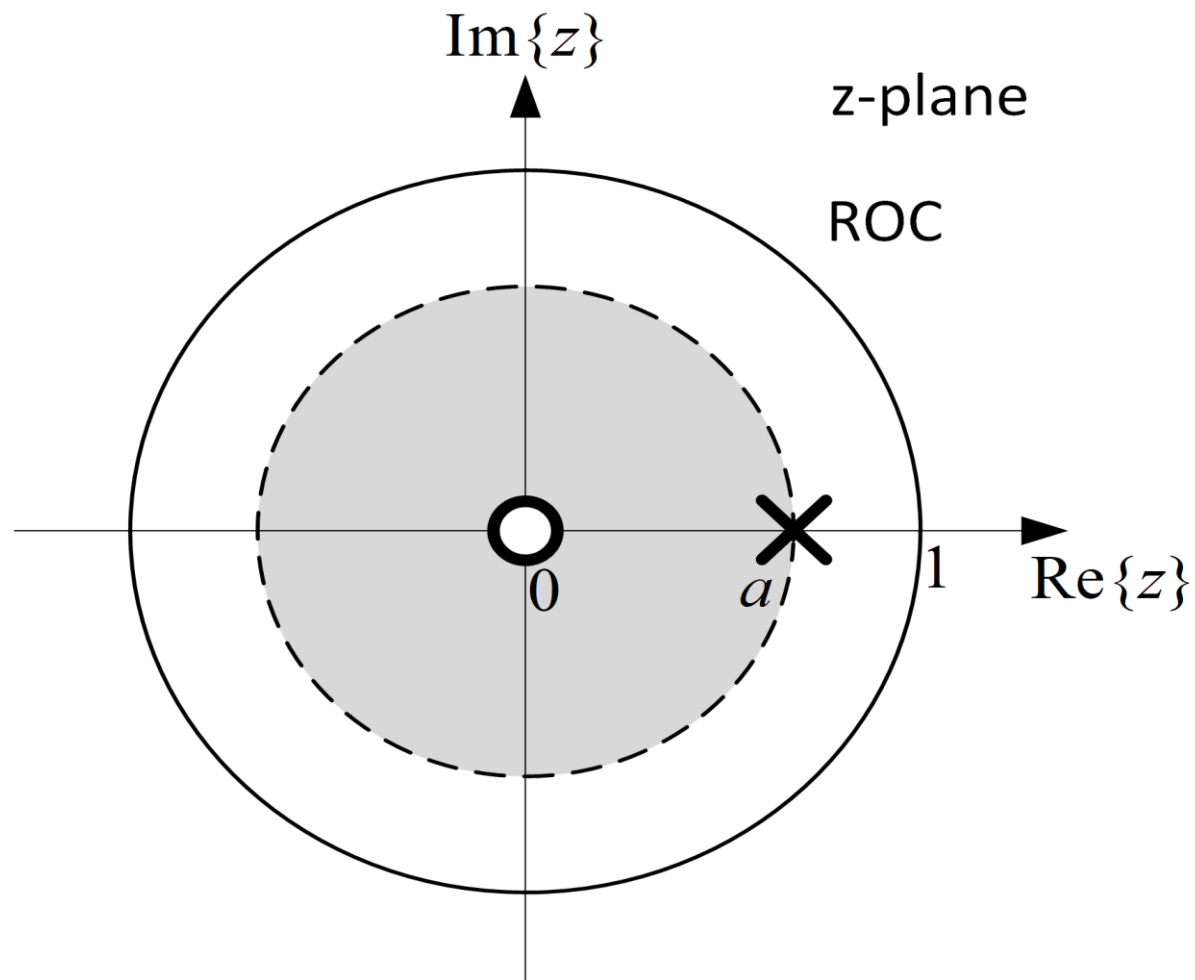
- Hence the solution of the z-transform is

$$X(z) = \frac{1}{1 - az^{-1}}, \quad |z| < |a|$$

- The z-transform pair is

$$-a^n u(-n - 1) \xleftrightarrow{z} \frac{1}{1 - az^{-1}}, \quad |z| < |a|$$

$$X(z) = \frac{1}{1 - az^{-1}}, \quad |z| < |a|$$



Example 4.4

- Determine the z-transform and its ROC of

$$x(n) = 7\left(\frac{1}{3}\right)^n u(n) - 6\left(\frac{1}{2}\right)^n u(n)$$

$$X(z) = \sum_{n=0}^{\infty} 7\left(\frac{1}{3}\right)^n z^{-n} - \sum_{n=0}^{\infty} 6\left(\frac{1}{2}\right)^n z^{-n}$$

$$= 7 \sum_{n=0}^{\infty} \left(\frac{1}{3} z^{-1}\right)^n - 6 \sum_{n=0}^{\infty} \left(\frac{1}{2} z^{-1}\right)^n$$

$$= \frac{7}{1 - \frac{1}{3} z^{-1}} - \frac{6}{1 - \frac{1}{2} z^{-1}}$$

$$= \frac{1 - \frac{3}{2} z^{-1}}{\left(1 - \frac{1}{3} z^{-1}\right)\left(1 - \frac{1}{2} z^{-1}\right)},$$

$$\underbrace{\left| \frac{1}{2} z^{-1} \right| < 1 \cap \left| \frac{1}{2} z^{-1} \right| < 1}_{=\text{ROC}}$$

The overall ROC is the intersection of ROC1 and ROC2

- With $\left| \frac{1}{3} z^{-1} \right| < 1 \cap \left| \frac{1}{2} z^{-1} \right| < 1$

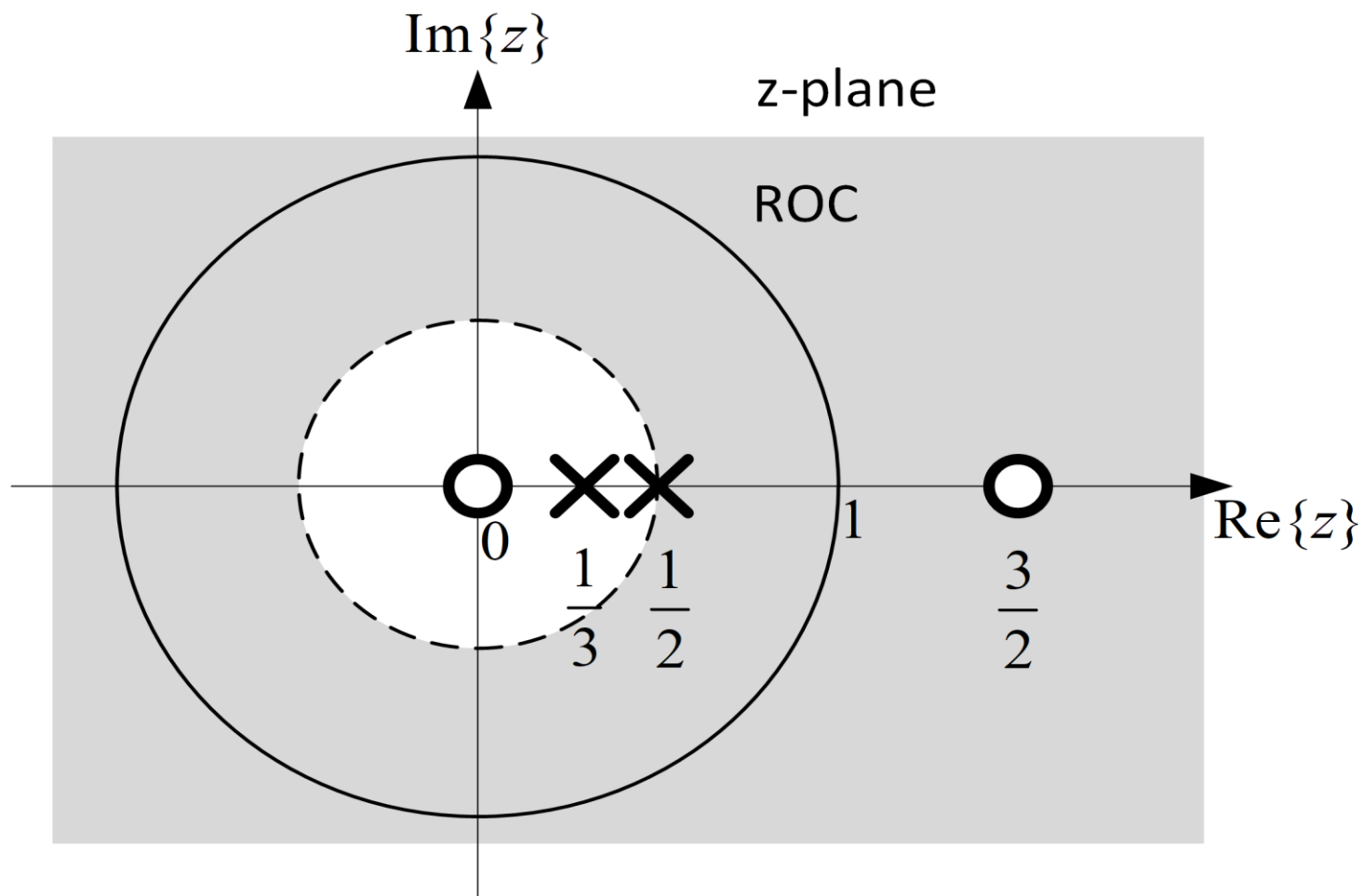
$$\left| \frac{1}{3} \right| < |z| \cap \left| \frac{1}{2} \right| < |z|$$

- Therefore,

$$|z| > \left| \frac{1}{2} \right|$$

$$X(z) = \frac{z(z - \frac{3}{2})}{(z - \frac{1}{3})(z - \frac{1}{2})}, \quad |z| > \left| \frac{1}{2} \right|$$

$$X(z) = \frac{z(z - \frac{3}{2})}{(z - \frac{1}{3})(z - \frac{1}{2})}, \quad |z| > \left| \frac{1}{2} \right|$$



Example 4.5

- Consider the z-transform of the unit impulse response

$$\delta(n) \xleftrightarrow{z} \sum_{n=-\infty}^{\infty} \delta(n) z^{-n} = 1$$

- The z-transform of the delayed unit impulse will be

The z-transform of delayed impulses

- The unit-delayed unit impulse

$$\delta(n-1) \xleftrightarrow{Z} \sum_{n=-\infty}^{\infty} \delta(n-1)z^{-n} = z^{-1}$$

The m-delayed unit impulse

$$\delta(n-m) \xleftrightarrow{Z} \sum_{n=-\infty}^{\infty} \delta(n-m)z^{-n} = z^{-m}$$

The Inverse z-Transform

- From
$$X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)(re^{j\omega})^{-n}$$
$$= \sum_{n=-\infty}^{\infty} (x(n)r^{-n})e^{-j\omega n}$$

We then have the inverse z-transform

$$x(n)r^{-n} = F^{-1}\{X(re^{j\omega})\}$$

$$x[n] = r^n F^{-1} \{X(re^{j\omega})\}$$

$$= r^n \frac{1}{2\pi} \int_{2\pi} X(re^{j\omega}) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{2\pi} X(re^{j\omega}) r^n e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{2\pi} X(re^{j\omega}) (re^{j\omega})^n d\omega$$

- With $z = re^{j\omega}$ and r is fixed

- $$dz = dre^{j\omega} = jre^{j\omega}d\omega$$

$$d\omega = \frac{1}{j} z^{-1} dz$$

- From
$$x(n) = \frac{1}{2\pi} \int_{2\pi} X(re^{j\omega}) (re^{j\omega})^n d\omega$$
$$= \frac{1}{2\pi} \oint X(z) z^n \left(\frac{1}{j} z^{-1} dz \right)$$
- We have the inverse z-transform as

$$x(n) = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$$

Important properties of the z-transform

- Shifting in the time-domain

$$Z[x(n - m)] = z^{-m} X(z)$$

- Convolution in the time-domain

$$y(n) = \sum_{k=-\infty}^{\infty} h(k)x(n - k) \Rightarrow Y(z) = H(z)X(z)$$

- Multiplying with n in the time domain

$$Z[nx(n)] = -z \frac{dX(z)}{dz}$$

$$Z[x(n-m)] = z^{-m} X(z)$$

$$\begin{aligned}
 X(z) &= \sum_{n=-\infty}^{\infty} x(n-m)z^{-n} \\
 &= \sum_{k+m=-\infty}^{\infty} x(k)z^{-(k+m)} \\
 &= \sum_{k=-\infty}^{\infty} x(k)z^{-k} \cdot z^{-m} \\
 &= z^{-m} \sum_{k=-\infty}^{\infty} x(k)z^{-k} \\
 &= z^{-m} X(z)
 \end{aligned}$$

$$Y(z) = Z\left[\sum_{k=-\infty}^{\infty} h(k)x(n-k)\right] = H(z)X(z)$$

$$\begin{aligned} Y(z) &= \sum_{n=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} h(k)x(n-k) \right] z^{-n} \\ &= \sum_{l+k=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} h(k)x(l) \right] z^{-(l+k)} \\ &= \sum_{l=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} h(k)x(l)z^{-k} \right] z^{-l} \\ &= \sum_{l=-\infty}^{\infty} \left[\sum_{k=-\infty}^{\infty} h(k)z^{-k} \right] x(l)z^{-l} \\ &= \sum_{l=-\infty}^{\infty} x(l)z^{-l} \cdot \sum_{k=-\infty}^{\infty} h(k)z^{-k} \\ &= X(z)H(z) \end{aligned}$$

$$x(n) = A\delta(n)$$

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} A\delta(n)z^{-n} \\ &= Az^{-0} \\ &= A \end{aligned}$$

$$X(z) = A$$

$$x(n) = Au(n)$$

$$X(z) = \sum_{n=-\infty}^{\infty} Au(n)z^{-n}$$

$$= \sum_{n=0}^{\infty} Az^{-n}$$

$$= \sum_{n=0}^{\infty} A(z^{-1})^n$$

$$= A \sum_{n=0}^{\infty} (z^{-1})^n$$

$$X(z) = \frac{A}{1 - z^{-1}}$$

$$x(n) = a^n u(n)$$

$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} a^n u(n) z^{-n} \\ &= \sum_{n=0}^{\infty} a^n z^{-n} \\ &= \sum_{n=0}^{\infty} (az^{-1})^n \\ &= \frac{1}{1 - az^{-1}} \end{aligned}$$

$a^n u(n) \xleftrightarrow{z} \frac{1}{1 - az^{-1}}$
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Important z-transform pairs

Table 1 z-transform pairs

Sequence $x(n)$	z-transform $X(z)$	Region of Convergence (ROC)
$\delta(n)$	1	All z
$u(n)$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
$a^n u(n)$	$\frac{1}{1 - az^{-1}}$	$ z > a $
$-a^n u(-n-1)$	$\frac{1}{1 - az^{-1}}$	$ z < a $
$2 C p ^n \cos(\omega_0 n + \theta)u(n)$	$\frac{C}{1 - pz^{-1}} + \frac{C^*}{1 - p^* z^{-1}}$	$ z > p $

Inversion of the z-transform

- Inverse z-transform

$$x(n) \equiv Z^{-1}[X(z)]$$

- Consider

$$X(z) = \frac{a_0 + a_1z + a_2z^2 + \dots + a_Nz^N}{b_0 + b_1z + b_2z^2 + \dots + b_Mz^M}$$

- Whose denominator can be written as

$$X(z) = \frac{a_0 + a_1z + a_2z^2 + \dots + a_Nz^N}{(z - p_1)(z - p_2)\dots(z - p_M)}$$

Three types for Poles

- The type of poles are categorised as:
 - 1) Poles are real numbers
 - 2) Poles are conjugate complex numbers
 - 3) Poles are repetitive numbers
- Partial Fraction Expansion (PFE) method is performed and then use Table 1 for inversion of the z-transform.

Partial Fraction Expansion (PFE)

- From

$$X(z) = \frac{a_0 + a_1z + a_2z^2 + \dots + a_Nz^N}{(z - p_1)(z - p_2)\dots(z - p_M)}$$

- PFE is used to find

$$X(z) = \frac{C_1}{(z - p_1)} + \frac{C_2}{(z - p_2)} + \dots + \frac{C_M}{(z - p_M)}$$

- Then consult Table 1 for $x(n)$ for each term.

Example 4.6 Real Distinct Poles

Determine the inverse z-transform of a causal system with

$$Y(z) = \frac{4z^2}{z^2 - 0.25}$$

Solution:

$$\begin{aligned} Y(z) &= \frac{4z^2}{z^2 - 0.25} \\ &= \frac{4z^2}{(z - 0.5)(z + 0.5)} = \frac{C_1 z}{z - 0.5} + \frac{C_2 z}{z + 0.5} \end{aligned}$$

Find C1 and C2

- Find C1
$$\frac{4z^2 \cancel{(z-0.5)}}{\cancel{(z-0.5)}(z+0.5)} \Big|_{z=0.5} = \frac{C_1 \cancel{z(z-0.5)}}{\cancel{(z-0.5)}} + \frac{C_2 z(z-0.5)}{(z+0.5)}$$
$$C_1 = \frac{4z}{z+0.5} \Big|_{z=0.5} = \frac{4(0.5)}{1} = 2$$
- Find C2
$$\frac{4z^2 \cancel{(z+0.5)}}{(z-0.5)\cancel{(z+0.5)}} \Big|_{z=-0.5} = \frac{C_1 z(z+0.5)}{(z-0.5)} + \frac{C_2 \cancel{z(z+0.5)}}{\cancel{(z+0.5)}}$$
$$C_2 = \frac{4z}{z-0.5} \Big|_{z=-0.5} = \frac{4(-0.5)}{-1} = 2$$

- We then have $Y(z) = \frac{2z}{z-0.5} + \frac{2z}{z+0.5}$
- or $Y(z) = \frac{2}{1-0.5z^{-1}} + \frac{2}{1+0.5z^{-1}}$

From Table 1 we have

$a^n u(n)$	$\frac{1}{1-az^{-1}}$	$ z > a $
$-a^n u(-n-1)$	$\frac{1}{1-az^{-1}}$	$ z < a $

Z-transform pair for causal systems

- Since the system is causal, we then choose

$a^n u(n)$	$\frac{1}{1 - az^{-1}}$	$ z > a $
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- Therefore

$$Y(z) = \frac{2}{1 - 0.5z^{-1}} + \frac{2}{1 + 0.5z^{-1}}$$

has an inverse

$$y(n) = 2(0.5)^n u(n) + 2(-0.5)^n u(n)$$

Homework 4.1

- Determine the inverse z-transform of $Y(z)$ from a causal system where

$$Y(z) = \frac{z^2}{z^2 + z - 2}$$

using the z-transform pair in Table 1.

Example 4.7 Complex Conjugate Poles

$$Y(z) = \frac{z}{z^2 + 1}$$

$$Y(z) = \frac{z}{z^2 + 1} = \frac{z}{(z - \sqrt{-1})(z + \sqrt{-1})}$$

$$Y(z) = \frac{C_1 z}{z - j} + \frac{C_2 z}{z + j}$$

Find C1

$$\frac{z(z-j)}{z^2+1} = \frac{C_1 z(z-j)}{z-j} + \frac{C_2 z(z-j)}{z+j}$$

$$\left. \frac{1}{(z+j)} \right|_{z=j} = C_1 + \left. \frac{C_2(z-j)}{z+j} \right|_{z=j}$$

$$C_1 = \left. \frac{1}{(z+j)} \right|_{z=j} = \frac{1}{(j+j)} = \frac{1}{2j} = -0.5j$$

Find C2

$$\frac{z(z+j)}{z^2+1} = \frac{C_1 z(z+j)}{z-j} + \frac{C_2 z(z+j)}{z+j}$$

$$\left. \frac{1}{(z-j)} \right|_{z=-j} = \left. \frac{C_1(z+j)}{z-j} \right|_{z=-j} + C_2$$

$$C_2 = \left. \frac{1}{(z-j)} \right|_{z=-j} = \frac{1}{(-j-j)} = \frac{-1}{2j} = 0.5j$$

$$Y(z) = \frac{-0.5jz}{z-j} + \frac{0.5jz}{z+j}$$

$$j = 0 + j$$

$$= \cos\left(\frac{\pi}{2}\right) + j \sin\left(\frac{\pi}{2}\right) = e^{j\frac{\pi}{2}}$$

$$-j = 0 - j$$

$$= \cos\left(-\frac{\pi}{2}\right) + j \sin\left(-\frac{\pi}{2}\right) = e^{-j\frac{\pi}{2}}$$

$$Y(z) = \frac{0.5e^{-j\frac{\pi}{2}}z}{z - e^{j\frac{\pi}{2}}} + \frac{0.5e^{j\frac{\pi}{2}}z}{z - e^{-j\frac{\pi}{2}}}$$

Diagram illustrating the partial fraction decomposition of $Y(z)$ into two terms. The first term is $\frac{0.5e^{-j\frac{\pi}{2}}z}{z - e^{j\frac{\pi}{2}}}$ and the second term is $\frac{0.5e^{j\frac{\pi}{2}}z}{z - e^{-j\frac{\pi}{2}}}$. The poles are marked with circles and labeled with their magnitude and angle: $0.5 \angle -\frac{\pi}{2}$ for the first pole and $1 \angle \frac{\pi}{2}$ for the second pole.

Then

$$Y(z) = \frac{0.5e^{-j\frac{\pi}{2}}z}{z - 1e^{j\frac{\pi}{2}}} + \frac{0.5e^{j\frac{\pi}{2}}z}{z - 1e^{-j\frac{\pi}{2}}}$$

Is written as

$$Y(z) = \frac{0.5e^{-j\frac{\pi}{2}}}{1 - 1e^{j\frac{\pi}{2}}z^{-1}} + \frac{0.5e^{j\frac{\pi}{2}}}{1 - 1e^{-j\frac{\pi}{2}}z^{-1}}$$

For

$$Y(z) = \frac{C}{1 - pz^{-1}} + \frac{C^*}{1 - p^* z^{-1}}$$

Using Table 1 for each term we have

$$y(n) = C(p)^n u(n) + C^* (p^*)^n u(n)$$

where $C = |C|e^{j\theta}$ and $p = |p|e^{j\omega_0}$

$$y(n) = (C(p)^n + C^* (p^*)^n)u(n)$$

By summing with its conjugate, we have real part

$$y(n) = 2 \operatorname{Re}(C(p)^n)u(n)$$

Then

$$y(n) = 2 \operatorname{Re}(|C| e^{j\theta} |p|^n e^{j\omega_0 n}) u(n)$$

So

$$y(n) = 2 \operatorname{Re}(|C| |p|^n e^{j(\omega_0 n + \theta)}) u(n)$$

From $\operatorname{Re}(\exp(jA)) = \cos(A)$

$$y(n) = 2 |C| |p|^n \cos(\omega_0 n + \theta) u(n)$$

Therefore

$$Y(z) = \frac{C}{1 - pz^{-1}} + \frac{C^*}{1 - p^*z^{-1}} \quad \begin{aligned} C &= |C|e^{j\theta} \\ p &= |p|e^{j\omega_0} \end{aligned}$$

has inverse z-transform of

$$y(n) = 2|C||p|^n \cos(\omega_0 n + \theta)u(n)$$

Hence

$$Y(z) = \frac{0.5e^{-j\frac{\pi}{2}}}{1 - 1e^{j\frac{\pi}{2}}z^{-1}} + \frac{0.5e^{j\frac{\pi}{2}}}{1 - 1e^{-j\frac{\pi}{2}}z^{-1}}$$

Has inverse z-transform of

$$\begin{aligned} y(n) &= 2|0.5||1|^n \cos\left(\frac{\pi}{2}n - \frac{\pi}{2}\right)u(n) \\ &= \cos\left(\frac{\pi}{2}n - \frac{\pi}{2}\right)u(n) \end{aligned}$$

Example 4.8 Repeated Real Poles

$$Y(z) = \frac{z^2}{(z - 0.5)(z - 1)^2}$$

$$Y(z) = \frac{z^2}{(z - 0.5)(z - 1)^2} = \frac{C_1 z}{(z - 0.5)} + \frac{C_2 z}{(z - 1)^2} + \frac{C_3 z}{z - 1}$$

$$\left. \frac{\cancel{z^2} \cancel{(z - 0.5)}}{\cancel{(z - 0.5)} (z - 1)^2} \right|_{z=0.5} = \frac{C_1 \cancel{z} \cancel{(z - 0.5)}}{\cancel{(z - 0.5)}} + \frac{C_2 \cancel{z} \cancel{(z - 0.5)}}{(z - 1)^2} + \frac{C_3 \cancel{z} \cancel{(z - 0.5)}}{z - 1}$$

$C_1 = \left. \frac{z}{(z - 1)^2} \right _{z=0.5} = \frac{0.5}{(0.5 - 1)^2} = 2$
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Find C2

$$\left. \frac{z^2 \cancel{(z-1)^2}}{(z-0.5) \cancel{(z-1)^2}} \right|_{z=1} = \frac{C_1 z \cancel{(z-1)^2}}{(z-0.5)} + \frac{C_2 z \cancel{(z-1)^2}}{\cancel{(z-1)^2}} + \frac{C_3 z \cancel{(z-1)^2}}{z-1}$$

$$C_2 = \left. \frac{z}{(z-0.5)} \right|_{z=1} = \frac{1}{(1-0.5)} = 2$$

Find C3

$$\left. \frac{z^2 \cancel{(z-1)}}{(z-0.5)(z-1)} \right|_{z=1} = \frac{C_1 z \cancel{(z-1)}}{(z-0.5)} + \frac{C_2 z \cancel{(z-1)}}{z-1} + \frac{C_3 z \cancel{(z-1)}}{\cancel{(z-1)}}$$

We cannot substitute $z=1$, then leave it for now.

Substitute $C_2=2$

Substitute $C_2=2$

$$\frac{z^2}{(z-0.5)(z-1)} \Big|_{z=1} = \frac{C_1 z(z-1)}{(z-0.5)} + \frac{(2)z}{z-1} + C_3 z$$

Find C_3

$$C_3 = \frac{z}{(z-0.5)(z-1)} \Big|_{z=1} - \frac{C_1(z-1)}{(z-0.5)} \Big|_{z=1} - \frac{(2)}{z-1} \Big|_{z=1}$$

$$C_3 = \frac{z - C_1(z-1)^2 - 2(z-0.5)}{(z-0.5)(z-1)} \Big|_{z=1}$$

$$= \frac{z - 2(z-0.5) - C_1(z-1)^2}{(z-0.5)(z-1)} \Big|_{z=1}$$

Now use $z=1$

$$= \frac{-\cancel{(z-1)} - C_1(z-1)}{(z-0.5)\cancel{(z-1)}} \Big|_{z=1}$$

$$= \frac{-1-0}{(1-0.5)} = -2$$

$$Y(z) = \frac{z^2}{(z - 0.5)(z - 1)^2} = \frac{2z}{(z - 0.5)} + \frac{2z}{(z - 1)^2} - \frac{2z}{z - 1}$$

$$y(n) = 2(0.5)^n + 2n - 2$$

Applications of the z-Transform

- 1. Find the impulse response of the system.
- 2. Find the convolution results.
- 3. Find the solution of difference equations.

Example 4.9: Find the impulse response of the following causal system

$$y(n) = 0.9y(n-1) + x(n)$$

Solution: Perform the z-transform

$$Y(z) = 0.9z^{-1}Y(z) + X(z)$$

$$\frac{Y(z)}{X(z)} = \frac{1}{1 - 0.9z^{-1}} = H(z)$$

Using Table 1

$$h(n) = (0.9)^n u(n)$$

Example 4.10: Find the convolution result of $h(n)$ and $x(n)$ where

$$h(n) = (0.5)^n u(n) \quad \text{and} \quad x(n) = \left(\frac{1}{3}\right)^n u(n)$$

Solution $Y(z) = H(z)X(z)$

$$H(z) = \frac{z}{z - 0.5}, \quad |z| > 0.5 \quad X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n} = \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n z^{-n}$$

$$X(z) = \sum_{n=0}^{\infty} \left(\frac{1}{3} z^{-1}\right)^n = \frac{1}{1 - \frac{1}{3} z^{-1}} = \frac{z}{z - \frac{1}{3}}, \quad |z| > \frac{1}{3}$$

$$Y(z) = \frac{z}{z - 0.5} \cdot \frac{z}{z - \frac{1}{3}}$$

$$Y(z) = \frac{z^2}{(z - 0.5)(z - \frac{1}{3})} = \frac{C_1 z}{(z - 0.5)} + \frac{C_2 z}{(z - \frac{1}{3})}$$

$$C_1 = 3, C_2 = -2 \quad \Longrightarrow \quad Y(z) = \frac{3z}{(z - 0.5)} - \frac{2z}{(z - \frac{1}{3})}$$

Use Table 1

$$y(n) = 3(0.5)^n u(n) - 2 \left(\frac{1}{3} \right)^n u(n)$$

Example 4.11: Find the solution of the following difference equation

$$y(n) - y(n-1) + 0.5y(n-2) = 0.5x(n) + 0.5x(n-1)$$

$y(n)$ is the angular position and input $x(n)$ is

$$x(n) = \delta(n)$$

Solution: The z-transform of $y(n)$ is

$$Y(z) - z^{-1}Y(z) + 0.5z^{-2}Y(z) = 0.5X(z) + 0.5z^{-1}X(z)$$

$$Y(z) = \frac{0.5 + 0.5z^{-1}}{1 - z^{-1} + 0.5z^{-2}} X(z)$$

From $Y(z) = H(z)X(z)$

We have $H(z) = \frac{0.5 + 0.5z^{-1}}{1 - z^{-1} + 0.5z^{-2}}$

Then

$$Y(z) = \frac{0.5 + 0.5z^{-1}}{1 - z^{-1} + 0.5z^{-2}} \cdot \frac{z^2}{z^2}$$

$$Y(z) = \frac{(0.5z + 0.5)z}{z^2 - z + 0.5}$$

$$Y(z) = \frac{(0.5z + 0.5)z}{(z - (0.5 + j0.5))(z - (0.5 - j0.5))}$$

$$Y(z) = \frac{C_1 z}{z - (0.5 + j0.5)} + \frac{C_2 z}{z - (0.5 - j0.5)}$$

$$\frac{(0.5z + 0.5)z}{(z - (0.5 + j0.5))(z - (0.5 - j0.5))} = \frac{C_1 z}{z - (0.5 + j0.5)} + \frac{C_2 z}{z - (0.5 - j0.5)}$$

Note that

$$(0.5 + j0.5) = 0.707 \angle 0.785 = 0.707 \angle \pi/4$$

and

$$(0.5 - j0.5) = 0.707 \angle -0.785 = 0.707 \angle -\pi/4$$

Find C1

$$\frac{(0.5z + 0.5)z}{(z - (0.5 - j0.5))} = C_1z + \frac{C_2z(z - (0.5 + j0.5))}{z - (0.5 - j0.5)}$$

$$z = 0.5 + j0.5$$

$$\frac{(0.5(0.5 + j0.5) + 0.5)z}{(0.5 + j0.5 - (0.5 - j0.5))} = C_1z$$

$$\frac{(0.75 + j0.25)}{j} = C_1$$

$$C_1 = 0.25 - j0.75 = 0.79 \angle -1.25$$

Find C2

$$\frac{(0.5z + 0.5)z}{(z - (0.5 + j0.5))} = \frac{C_1 z(z - (0.5 - j0.5))}{z - (0.5 + j0.5)} + C_2 z$$

$$z = 0.5 - j0.5$$

$$\frac{(0.5(0.5 - j0.5) + 0.5)z}{(0.5 - j0.5 - (0.5 + j0.5))} = C_1 z$$

$$\frac{(0.75 - j0.25)}{-j} = C_2$$

$$C_2 = 0.25 + j0.75 = 0.79 \angle 1.25$$

$$Y(z) = \frac{(0.79 \angle -1.25)z}{z - 0.707 \angle \pi/4} + \frac{(0.79 \angle 1.25)z}{z - 0.707 \angle -\pi/4}$$

$$= \frac{0.79e^{-j1.25}z}{z - 0.707e^{j\pi/4}} + \frac{0.79e^{j1.25}z}{z - 0.707e^{-j\pi/4}}$$

From Table 1

$2 C p ^n \cos(\omega_0 n + \theta)u(n)$	$\frac{C}{1 - pz^{-1}} + \frac{C^*}{1 - p^* z^{-1}}$	$ z > p $
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$$y(n) = 1.58(0.707)^n \cos(\pi n / 4 - 1.25)u(n), \quad n \geq 0$$

Transfer function

Definition 1 $H(z)$ is called Transfer function
where

$$H(z) = \sum_{n=-\infty}^{\infty} h(n)z^{-n},$$

$$Y(z) = H(z)X(z) : \quad \text{ROC}_y = \text{ROC}_h \cap \text{ROC}_x$$

$$y(n) + \sum_{k=1}^N a_k y(n-k) = \sum_{l=1}^M b_l y(n-l)$$

$$Y(z) + \sum_{k=1}^N a_k z^{-k} Y(z) = \sum_{l=1}^M b_l z^{-l} X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{l=1}^M b_l z^{-l}}{1 + \sum_{k=1}^N a_k z^{-k}} = \frac{B(z)}{A(z)}$$

$$= \frac{b_0 z^{-M} \left(z^M + b_1 z^{(M-1)} + \dots + \frac{b_M}{b_0} \right)}{z^{-N} \left(z^N + a_1 z^{(N-1)} + \dots + a_N \right)}$$

$$= b_0 z^{N-M} \frac{(z - z_1)(z - z_2) \dots (z - z_M)}{(z - p_1)(z - p_2) \dots (z - p_N)}$$

We have the general form of a transfer function

$$H(z) = b_0 z^{N-M} \frac{\prod_{l=1}^M (z - z_l)}{\prod_{k=1}^N (z - p_k)}$$

Frequency Response from z-transform

We can find frequency response from the z-transform if its ROC includes the unit circle.

Transfer function

$$H(e^{j\omega}) = b_0 e^{j(N-M)\omega} \frac{\prod_{l=1}^M (e^{j\omega} - z_l)}{\prod_{k=1}^N (e^{j\omega} - p_k)} \bigg|_{z=e^{j\omega}}$$

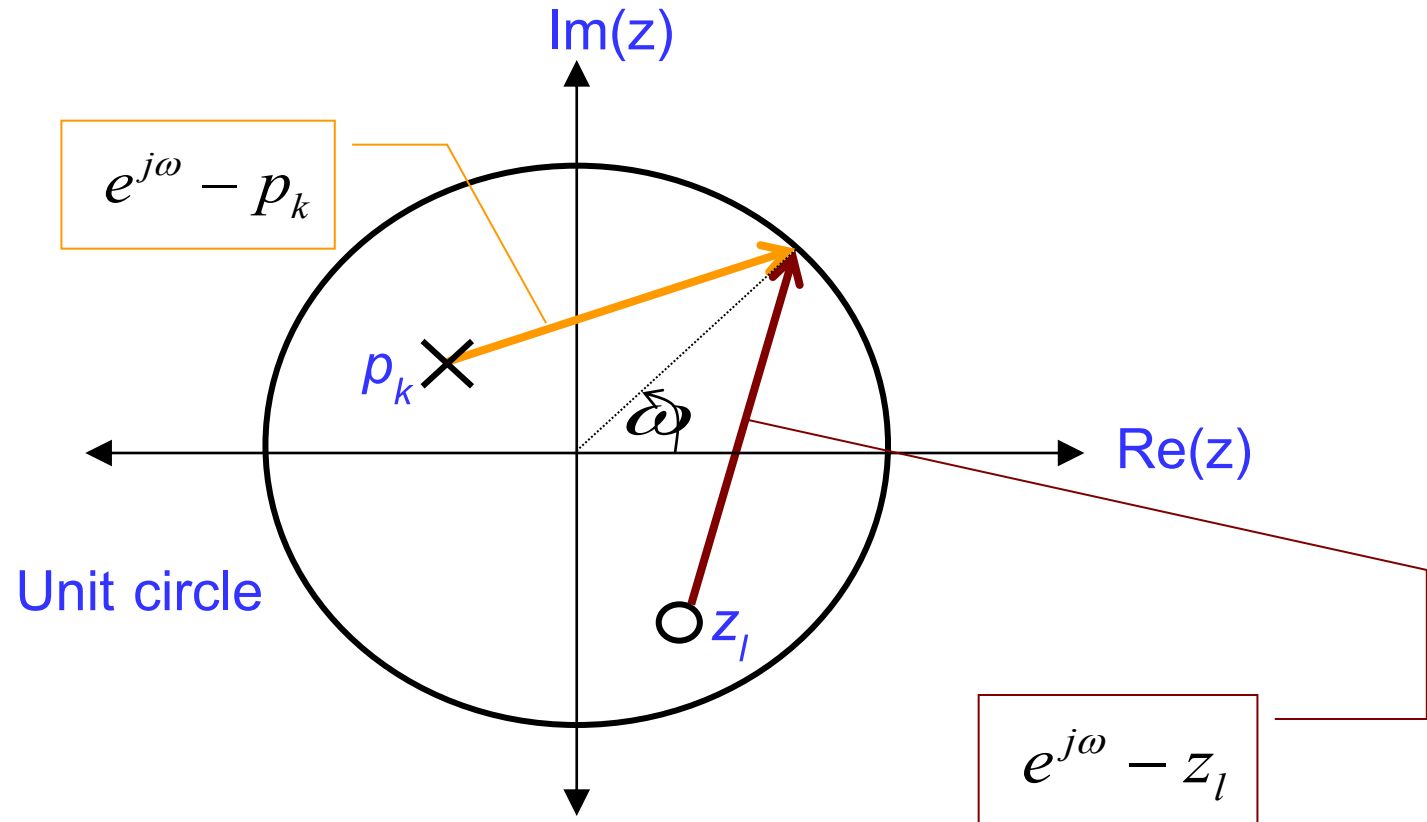
Magnitude response

$$|H(e^{j\omega})| = |b_0| \frac{|e^{j\omega} - z_1| \dots |e^{j\omega} - z_M|}{|e^{j\omega} - p_1| \dots |e^{j\omega} - p_N|}$$

Phase response

$$\angle H(e^{j\omega}) = \underbrace{[0 \text{ or } \pi]}_{\text{constant}} + \underbrace{\lceil (N-M)\omega \rceil}_{\text{linear}} + \underbrace{\sum \angle(e^{j\omega} - z_k) - \sum \angle(e^{j\omega} - p_k)}_{\text{nonlinear}}$$

Vectors from Poles and Zeros to the unit circle at a particular frequency



Example 4.12: Determine the transfer function and the frequency response of the following system

$$y(n) = 0.9y(n-1) + x(n)$$

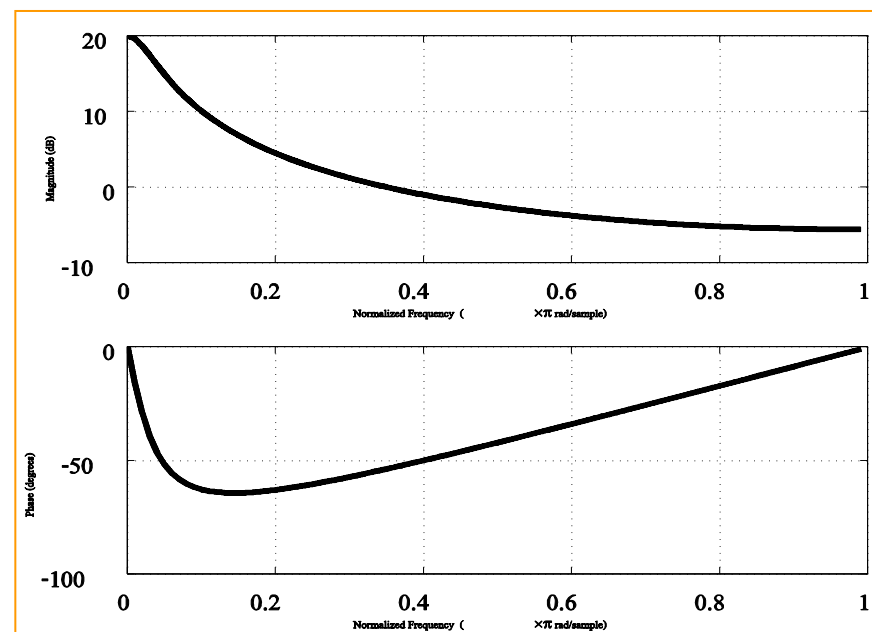
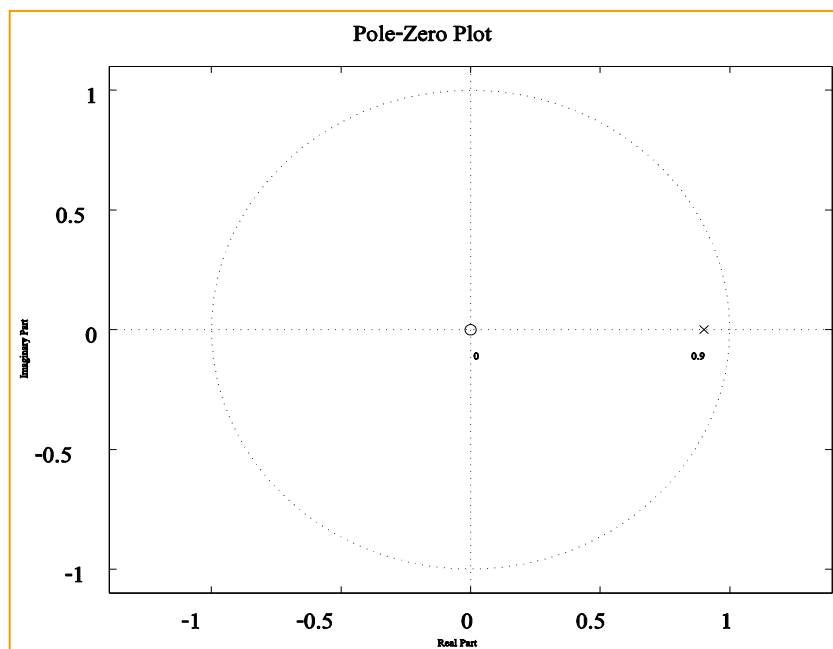
- Solution

The transfer function is

$$H(z) = \frac{1}{1 - 0.9z^{-1}}, \quad |z| > 0.9$$

The frequency response is

$$H(e^{j\omega}) = \frac{1}{1 - 0.9e^{-j\omega}} = \frac{e^{j\omega}}{e^{j\omega} - 0.9}$$



Homework 4.2

By means of the z-transform, determine the impulse response $h(n)$ of the following system

$$y(n) = 0.25y(n-2) + x(n)$$